

Messages > Building with Claude

Features overview

Explore Claude’s advanced features and capabilities.

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Claude’s API surface is organized into five areas:

- **Model capabilities:** Control how Claude reasons and formats responses.
- **Tools:** Let Claude take actions on the web or in your environment.
- **Tool infrastructure:** Handles discovery and orchestration at scale.
- **Context management:** Keeps long-running sessions efficient.
- **Files and assets:** Manage the documents and data you provide to Claude.

If you’re new, start with [model capabilities](#) and [tools](#). Return to the other sections when you’re ready to optimize cost, latency, or scale.

For administration and governance, see the [Admin API](#), the [Usage and Cost API](#), and the [Compliance API](#).

Feature availability

The Availability column in each of the following tables lists the platforms that offer a feature. A platform listed without a label offers the feature as stable, fully supported, and recommended for production use, with no beta header and with standard API [versioning](#) guarantees. A label after a platform name marks one of the following classifications on that platform. Not all features pass through every stage, and a feature may enter at any stage or skip stages.

Classification	Description
Classific	Preview features used for gathering feedback and iterating on a less mature use case. Availability may be limited, including through sign-up requirements or waitlists, and may not be publicly announced.
Beta	Features may change significantly or be discontinued based on feedback. Not guaranteed for ongoing production use. Breaking changes are possible with notice, and some platform-specific limitations may apply. Beta features on the Claude API and Claude Platform on AWS have a beta header .

**Deprecat
ed**

Feature is still functional but no longer recommended. A migration path and removal timeline are provided.

**Retir
ed**

Feature is no longer available.

Platform labels: Claude API (Anthropic first-party) · [Bedrock](#) (AWS-operated) · [Claude Platform on AWS](#) (Anthropic-operated on AWS) · [Google Cloud](#) (Google-operated) · [Microsoft Foundry](#) (Anthropic-operated on Azure)

Model capabilities

Ways to steer Claude and Claude's direct outputs, including response format, reasoning depth, and input modalities.

💡 You can discover which capabilities a model supports programmatically. The [Models API](#) returns `max_input_tokens`, `max_tokens`, and a `capabilities` object for every available model.

The ZDR column indicates whether a feature is available under a Zero Data Retention arrangement. For most features this depends only on what the feature mechanism retains; for features tied to specific models, model-level ZDR availability also applies. See [Model-specific data retention requirements](#).

Feature		Availability (ZDR)	Availability
<u>Context windows</u>	Up to 1M tokens for processing large documents, extensive code bases, and long conversations.	ZDR eligible	Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
<u>Adaptive thinking</u>	Let Claude dynamically decide when and how much to think. The only thinking mode on Claude 4.7 and later models. Use the effort parameter to control thinking depth.	ZDR eligible	Microsoft Foundry
			Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
			Microsoft Foundry

Feature	Description	Availability (ZDR)	Availability
Batch processing	Process large volumes of requests asynchronously for cost savings. Send batches with a large number of queries per batch. Batch API calls cost 50% less than standard API calls.	Not ZDR eligible	Claude API Claude Platform on AWS
Citations	Ground Claude's responses in source documents. With Citations, Claude can provide detailed references to the exact sentences and passages it uses to generate responses, leading to more verifiable, trustworthy outputs.	ZDR eligible	Claude API Bedrock Claude Platform on AWS Google Cloud Microsoft Foundry
Data residency	Control where model inference runs using geographic controls. Specify "global" or "us" routing per request through the inference_geo parameter.	ZDR eligible	Claude API Claude Platform on AWS

Parameter	Description	Availability (ZDR)	Availability
<u>Effort</u>	Control how many tokens Claude uses when responding with the effort parameter, trading off between response thoroughness and token efficiency.	ZDR eligible	Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
<u>Fallback credit</u>	Avoid paying the prompt-cache cost twice when you retry a refused request on another model. The refusal carries a credit token, and echoing it on the retry bills the retry as though the conversation had been on the new model all along. Message Batches results do not include fallback credit tokens.	Not ZDR eligible*	Microsoft Foundry
			Claude API (Beta)
			Bedrock (Beta)
			Claude Platform on AWS (Beta)
			Google Cloud (Beta)
			Microsoft Foundry (Beta)

Feature	Description	Integration (ZDR)	Availability
PDF support	Process and analyze text and visual content from PDF documents.	ZDR eligible	Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
Search results	Enable natural citations for RAG applications by providing search results with proper source attribution. Achieve web search-quality citations for custom knowledge bases and tools.	ZDR eligible	Microsoft Foundry
			Claude API
			Bedrock
			Claude Platform on AWS
Server-side fallback	Retry a refused request inside a single API call. Use the "default" mode to apply Anthropic's recommended fallback models, or name up to three models of your own; when the requested model declines, the API runs the next model in the chain on the same request. The fallbacks parameter is not available in the Message Batches API.	Not ZDR eligible*	Google Cloud
			Microsoft Foundry
			Claude API (Beta)

Feature		Integration (ZDR)	Availability
			Claude API
			Bedrock
Structured outputs	Guarantee schema conformance with two approaches: JSON outputs for structured data responses, and strict tool use for validated tool inputs.	ZDR eligible (qualified)*	Claude Platform on AWS Google Cloud Microsoft Foundry
Thinking	Enhanced reasoning capabilities for complex tasks, providing transparency into Claude's step-by-step thought process before delivering its final answer.	ZDR eligible	Claude Platform on AWS Google Cloud Microsoft Foundry

Tools

Built-in tools that Claude invokes through `tool_use`. Server-side tools are run by the platform; client-side tools are implemented and executed by you.

Feature	Description	ZDR	Availability
<u>Advisor tool</u>	Pair a faster executor model with a higher-intelligence advisor model that provides strategic guidance mid-generation for long-horizon agentic workloads.	ZDR eligible	Claude API (Beta) Claude Platform on AWS (Beta)
<u>Code execution</u>	Run code in a sandboxed environment for advanced data analysis, calculations, and file processing. Free when used with web search or web fetch.	Not ZDR eligible	Claude API Claude Platform on AWS Microsoft Foundry†
<u>Web fetch</u>	Retrieve full content from specified web pages and PDF documents for in-depth analysis.	ZDR eligible *	Claude API Claude Platform on AWS Microsoft Foundry
<u>Web search</u>	Augment Claude's comprehensive knowledge with current, real-world data from across the web.	ZDR eligible *	Claude API Claude Platform on AWS Google Cloud Microsoft Foundry

Feature	Description	ZDR	Availability
<u>Bash</u>	Execute bash commands and scripts to interact with the system shell and perform command-line operations.	ZDR eligible	Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
<u>Browser use</u>	Navigate, read, and interact with webpages in your own browser environment.	ZDR eligible	Microsoft Foundry
			Claude API
<u>Computer use</u>	Control computer interfaces by taking screenshots and issuing mouse and keyboard commands.	ZDR eligible	Google Cloud
			Claude API
			Bedrock (Beta)
			Claude Platform on AWS (Beta)

<u>Memo ry</u>	Enable Claude to store and retrieve information across conversations. Build knowledge bases over time, maintain project context, and learn from past interactions.	ZDR eligible	Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
<u>Text editor</u>	Create and edit text files with a built-in text editor interface for file manipulation tasks.	ZDR eligible	Microsoft Foundry
			Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
			Microsoft Foundry

Tool infrastructure

Infrastructure that supports discovering, orchestrating, and scaling tool use.

Feature	Description	ZDR	Availability
<u>Agent Skills</u>	Extend Claude's capabilities with Skills. Use pre-built Skills (PowerPoint, Excel, Word, PDF) or create custom Skills with instructions and scripts. Skills use progressive disclosure to efficiently manage context.	Not ZDR eligible	Claude API
			Claude Platform on AWS (Beta)
			Microsoft Foundry (Beta) [†]

<u>Fine-grained tool streaming</u>	Stream tool use parameters without buffering/JSON validation, reducing latency for receiving large parameters.	ZDR eligible	Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
			Microsoft Foundry
<u>MCP connector</u>	Connect to remote <u>MCP</u> servers directly from the Messages API without a separate MCP client.	Not ZDR eligible	Claude API (Beta)
			Claude Platform on AWS (Beta)
			Microsoft Foundry (Beta)
<u>Programmatic tool calling</u>	Enable Claude to call your tools programmatically from within code execution containers, reducing latency and token consumption for multi-tool workflows.	Not ZDR eligible	Claude API
			Claude Platform on AWS
			Microsoft Foundry+

Tool search	Scale to thousands of tools by dynamically discovering and loading tools on-demand using regex- and BM25-based search, optimizing context usage and improving tool selection accuracy.	ZDR eligible	Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
			Microsoft Foundry

Context management

Infrastructure for controlling and optimizing Claude’s context window.

Feature	Description	ZDR	Availability
Compacti on	Server-side context summarization for long-running conversations. When context approaches the window limit, the API automatically summarizes earlier parts of the conversation.	ZDR eligible	Claude API (Beta)
			Bedrock (Beta)
			Claude Platform on AWS (Beta)
			Google Cloud (Beta)
			Microsoft Foundry (Beta)

Context editing	Automatically manage conversation context with configurable strategies. Supports clearing tool results when approaching token limits and managing thinking blocks in extended thinking conversations.	ZDR eligible	(Beta)
			Bedrock (Beta)
			Claude Platform on AWS (Beta)
			Google Cloud (Beta)
			Microsoft Foundry (Beta)
Automatic prompt caching	Simplify prompt caching to a single API parameter. The system automatically caches the last cacheable block in your request, moving the cache point forward as conversations grow.	ZDR eligible	Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
			Microsoft Foundry
Prompt caching (5m)	Provide Claude with more background knowledge and example outputs to reduce costs and latency.	ZDR eligible	Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
			Microsoft Foundry

<u>Prompt caching (1hr)</u>	Extended 1-hour cache duration for less frequently accessed but important context, complementing the standard 5-minute cache.	ZDR eligible	Bedrock
			Claude Platform on AWS
			Google Cloud
			Microsoft Foundry
<u>Token counting</u>	Token counting enables you to determine the number of tokens in a message before sending it to Claude, helping you make informed decisions about your prompts and usage.	ZDR eligible	Claude API
			Bedrock
			Claude Platform on AWS
			Google Cloud
			Microsoft Foundry

Files and assets

Manage files and assets for use with Claude.

Feature	Description	ZDR	Availability
<u>Files API</u>	Upload and manage files to use with Claude without re-uploading content with each request. Supports PDFs, images, and text files.	Not ZDR eligible	Claude API
			Claude Platform on AWS (Beta)
			Microsoft Foundry (Beta) [†]

message content, but both handle refusals from Claude Fable 5, which is not available under ZDR. See ZDR details.

† On Microsoft Foundry, feature availability differs by hosting option. These features are available on Hosted on Anthropic deployments, and not on Hosted on Azure deployments.

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Claude Platform Docs



Solutions

- AI agents
- Code modernization
- Coding
- Customer support
- Financial services
- Government
- Higher education
- K-12 teachers
- Life sciences

Partners

- Claude on AWS
- Claude on Google Cloud

Learn

- Blog
- Courses
- Use cases
- Connectors
- Customer stories
- Engineering at Anthropic
- Events
- Powered by Claude
- Service partners
- Startups program

Company

- Anthropic
- Careers
- Economic Futures
- Research

Security and compliance

Transparency

Help and security

Availability

Status

Support

Discord

Terms and policies

Privacy policy

Responsible disclosure policy

Terms of service: Commercial

Terms of service: Consumer

Usage policy